

ARAVINTH K

Senior Embedded Firmware Engineer

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PROFILE SUMMARY

Senior Embedded Firmware Engineer with 5+ years of experience in embedded firmware development, debugging, integration and validation across industrial automation and consumer electronics domains. Skilled in Embedded C/C++, Zephyr RTOS, Embedded Linux and communication protocols including CAN, SPI, I2C, UART and MQTT. Delivered firmware for three commercial Corsair sim-racing products and received several Bosch awards for quality and delivery excellence.

TECHNICAL SKILLS

Programming Languages: Embedded C, C++, Shell Scripting

RTOS & Operating Systems: Zephyr RTOS, Embedded Linux

Microcontrollers: STM32, TI TIVA C, NXP LPC Series, Atmel AVR/SAM, Microchip PIC

Communication Protocols: CAN, UART, SPI, I2C, RS232, RS485, MQTT

Wireless & Connectivity: Bluetooth, Wi-Fi, GPS, GSM, RFID

Embedded Linux: System Programming, POSIX Threads, Inter-Process Communication (IPC), Character Device Drivers

Embedded Systems: Firmware Development, Firmware Validation, Embedded Software Design, System Integration, Debugging, Root Cause Analysis

Testing & Quality: Unit Testing (Ceedling), Integration Testing, System Testing, Static Code Analysis (Cppcheck)

Tools & IDEs: Git, Gerrit, Version control, JIRA, GDB, JTAG, Oscilloscope, Logic Analyzer, VS Code, Keil MDK, MPLAB X, Atmel Studio, Code Composer Studio (CCS)

WORK EXPERIENCE

Analyst (Senior Embedded Firmware Engineer) | Bosch Global Software Technologies Pvt. Ltd. Feb 2023 – Present
Coimbatore, India

- Developed firmware for 3 commercial Corsair sim-racing products using Zephyr RTOS, including Handbrake, PVGT Porsche and Formula V3 Steering Wheel.
- Resolved 21+ defects across 4 Corsair product lines through defect resolution, root cause analysis, regression testing, as well as feature enhancements, improving product stability and reliability.
- Led and mentored a team of 2 junior firmware engineers, providing technical guidance, task planning, code reviews, and project onboarding support.

Embedded Software Developer | Kalycito
Coimbatore, India

Apr 2022 – Jan 2023

- Supported C++ application software on Linux for EdgeXConnect, a configurable no-code OPC UA Server platform for industrial IoT environments.
- Migrated the OPC UA server implementation from the licensed Matrikon OPC UA SDK to the open source open62541 library, eliminating third-party licensing costs while maintaining application functionality.
- Improved code organization and maintainability by restructuring the project using Git submodules and supporting integration testing and release validation.

- Sole firmware engineer for a 3-Layer Conveyor System (textile automation) delivered firmware development, system integration, testing and on-site deployment end-to-end in under 4 months.
- Calibrated and fine-tuned stepper motor control firmware for an Automatic Cone Winding Machine, implementing bug fixes and enhancements to improve automation performance and system reliability.
- Designed and validated a pulse oximeter proof-of-concept using the MAX30100 sensor, implementing SpO₂ and heart-rate measurement algorithms with $\pm 2\%$ accuracy.

KEY PROJECTS

Corsair Sustainability, PVGT Porsche & Formula V3 Steering Wheels

Bosch · 2024 – Present

Stack: *Embedded C, Zephyr RTOS, NXP LPC, CAN, SPI, I2C*

- Delivered sustainability support for multiple Corsair (Endor) sim-racing products through defect resolution, root cause analysis, regression testing and firmware enhancements.
- Developed and customized firmware for the PVGT Porsche and Formula V3 Steering Wheels using Zephyr RTOS by extending an existing firmware framework. Implemented product-specific features, performed product integration, system-level testing and supported release activities.

Corsair Handbrake — Sim Racing Peripheral Firmware

Bosch · 2023 - 2024

Stack: *Embedded C, Zephyr RTOS, NXP LPC, CAN Bus, Sensor Interface, EEPROM*

- Developed Zephyr RTOS firmware for a commercial sim-racing handbrake, implementing HX717 sensor acquisition, CAN communication, EEPROM-based profile storage and bootloader integration.
- Coordinated product release activities, including wheelbase compatibility validation, regression testing, and launch sign-off.

EdgeXConnect — Industrial No-Code OPC UA Server

Kalycito · 2022 – 2023

Stack: *C++, Linux, OPC UA SDK, CMake*

- Implemented the OPC UA server address space using the open62541 API by creating nodes, variables, methods, and object hierarchies based on application requirements.
- Successfully migrated the server implementation from the commercial Matrikon OPC UA SDK to the open source open62541 library while preserving existing server functionality.

AWARDS & RECOGNITION

Best Tyro — Bosch

Recognized for quick adaptation, strong commitment, and high productivity within a short ramp-up period.

Catalyst — Bosch

Excellent defect identification in Corsair Steering Wheel firmware ahead of the product launch schedule.

Growth Driver — Bosch

Outstanding contribution to the Corsair firmware sustainability initiative.

EDUCATION

B.E — Electronics and Communication Engineering

2013 – 2017

Nandha College of Technology, Erode